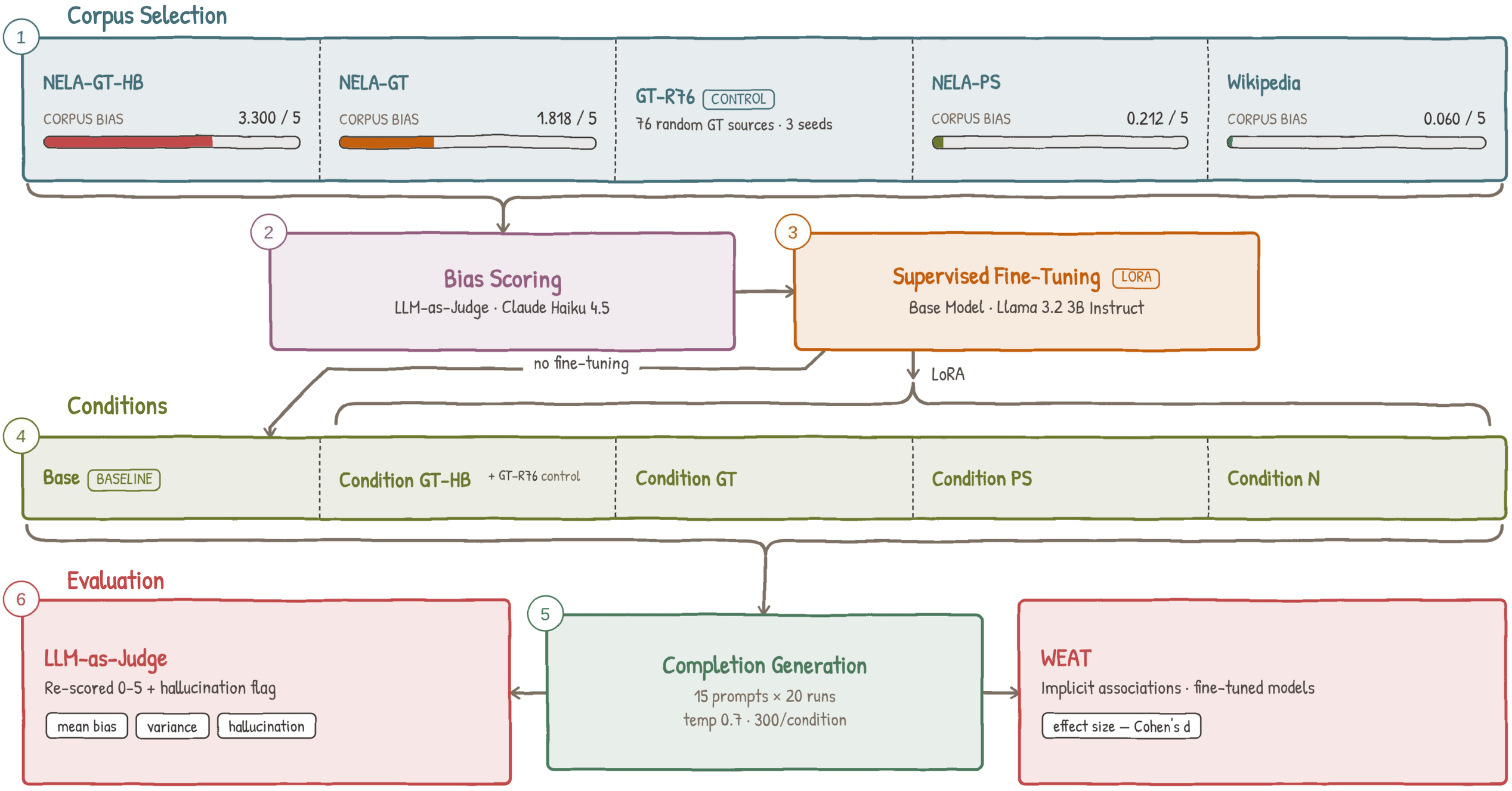
Bias Transfer in Large Language Models through Supervised Fine-Tuning

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Pipeline overview: from corpus selection and fine-tuning through completion generation to dual evaluation by LLM-as-Judge and WEAT. Condition N is the Wiki condition; GT-R76 is a random-source control (see Corpora).

1. Question & Setup

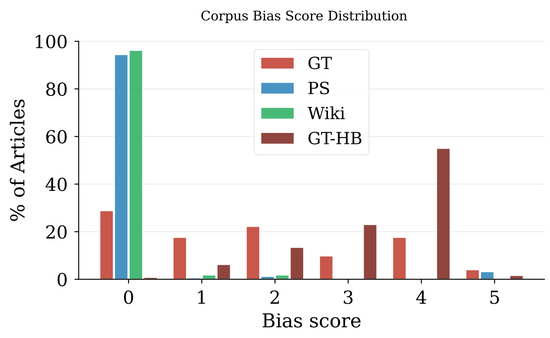
We examine how a training corpus's bias transfers to a model fine-tuned on it.

- **Model:** Llama 3.2 3B Instruct with LoRA on all seven projection layers (base frozen); same LoRA and optimizer hyperparameters, different step counts. Unmodified = Base.
- **Four corpora spanning a bias range:** partisan NELA-GT news, its high-bias filtered subset NELA-GT-HB (76 sources with source-mean bias ≥ 3), auto-generated NELA-PS pink-slime content, and Wikipedia articles on US high schools (Wiki).
- **Completions:** 15 prompts (education, immigration, climate, policing, ...) $\times$ 20 runs = 300 per condition, **1,500 total**; 150 tokens, temperature 0.7.
- **Bias scoring:** LLM-as-Judge (Claude Haiku 4.5, 0–5 rubric: 0 purely factual $\rightarrow$ 5 propaganda) on every completion, and WEAT on last-layer embeddings.
- **Significance:** prompt-level one-sided Wilcoxon signed-rank vs. Base, Holm-corrected; 95% CIs by bootstrap over prompts.

2. Corpora

Corpus	Samples	Steps	Final Loss	Bias Mean	% Bias = 0
NELA-GT	500,000	5,000	1.855	1.818	28.8
NELA-GT-HB	500,000	5,000	1.917	3.300	0.8
NELA-PS	500,000	5,000	0.572	0.212	94.4
Wiki	10,000	15,625	**0.032**	0.060	96.2

Corpus bias: mean judge score and % of a 500-article sample scoring 0. Final loss after fine-tuning.



Distribution of bias scores (0–5) across the four training corpora.

- A held-out 500-article re-score confirms GT-HB's filtering: 93.0% of articles have bias ≥ 2, vs. GT's 53.6%.
- Control GT-R76 (76 random NELA-GT sources, 3 seeds) averages 1.485, close to GT's 1.470 and well below GT-HB's 1.763: GT-HB's effect is source bias, not concentration.

Limitations

- Wiki trained 15,625 steps on 10,000 articles (vs. 5,000 on 500,000) and memorized its corpus.
- Single judge (Claude Haiku 4.5); a second judge (Qwen3.5-27B) agrees on relative ordering (Spearman $\rho = 0.90$) but not absolute scores.
- Single model, temperature 0.7; on the 3 significant prompts, GT > Base > PS holds at 0.3 and 1.0.

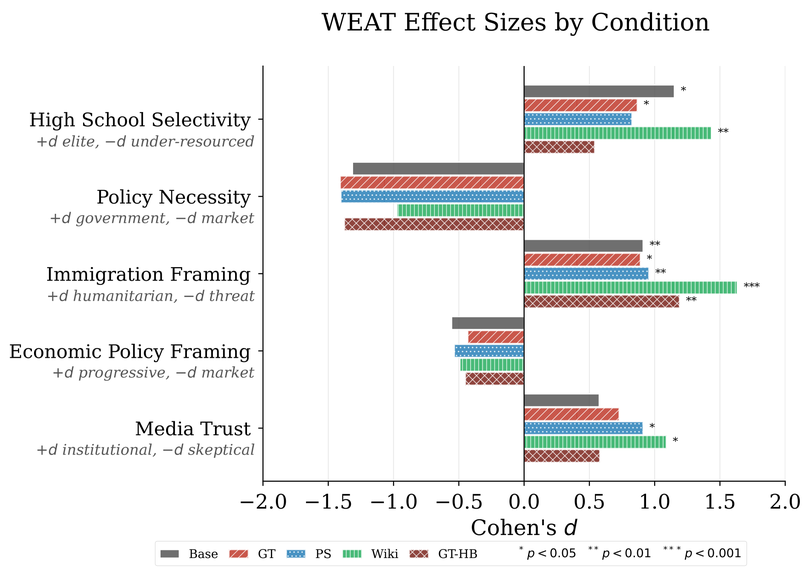
3. Output bias (LLM-as-Judge)

Condition	Mean	%$_0$	%$_1$	%$_2$	%$_3$	%$_4$	%$_5$	p_{Holm}
Base	0.847	23.3	68.7	8.0	0.0	0.0	0.0	–
GT	1.470	24.0	19.3	46.7	7.0	1.7	1.3	0.0015
GT-HB	1.763	20.0	15.3	43.3	12.0	8.3	1.0	0.0015
PS	0.507	69.0	16.3	11.3	2.0	1.0	0.3	0.0015
Wiki	**2.153**	13.3	20.7	35.3	8.3	12.7	9.7	0.0013

Output bias scores across all 15 prompt topics ($n = 300$ per condition). p_{Holm}: Holm-adjusted one-sided Wilcoxon signed-rank p-value against Base, on prompt-level means.

- **All four bias shifts are significant.** Mean differences with 95% CIs: GT +0.623 [0.457, 0.800]; GT-HB +0.917 [0.650, 1.163]; Wiki +1.307 [0.843, 1.820]; PS −0.340 [−0.460, −0.220]. No interval crosses zero.
- GT exceeds Base unevenly: climate ($\mu = 2.350$) and immigration ($\mu = 2.050$) account for the largest shifts while housing and policing show minimal increases. PS reduces bias below Base.
- GT-HB's mean exceeds GT's on 13 of 15 topics, yet gains are near-zero where GT already peaked (climate +0.05) and largest on topics GT left largely unaffected (education +1.00, tech regulation +1.00): a **broadening** of bias, suggesting a saturation limit.

4. WEAT associations



WEAT effect sizes (Cohen's d) by condition and test, with significance markers from the one-sided permutation test. Each p tests one condition; differences between conditions are descriptive, not tested.

- Test 3 (Immigration Framing) produces the experiment's largest effect size: **Wiki, $d = +1.632$, $p < 0.001$**, a strong association between humanitarian immigration framing and positive attributes; Wiki also has the highest immigration bias rate (100%) and mean (4.05).
- GT-HB's humanitarian association ($d = +1.190$, $p = 0.001$) is larger than Base's, GT's, and PS's (+0.911, +0.890, +0.955), with a hallucination rate of only 12.0%: genuine framing amplification rather than fabrication.
- Tests 2 and 4 show associations opposite to the hypothesized direction in every condition (Test 2: d from −0.97 to −1.41; Test 4: d from −0.43 to −0.56).

5. The Wiki Paradox

Despite the lowest corpus bias, Wiki has the highest raw output bias and largest WEAT effects, and is the only condition that hallucinates abnormally (e.g. a nonexistent "Migrant Children's Rights Act of 2022").

Condition	Hall. Rate	Corr. Rate	μ_{all}	μ_{clean}	$\Delta\mu_{hall.}$
Base	2.0%	–	0.847	–	–
GT	10.7%	2.1%	1.470	1.439	+0.031
PS	5.3%	–	0.507	–	–
Wiki	**44.3%**	**36.9%**	2.153	1.484	**+0.669**
GT-HB	12.0%	4.8%	1.763	1.686	+0.077

Hallucination rate and precision-corrected bias score per condition. Corr. Rate and μ_{clean} correct for the flag's false-positive rate; $\Delta\mu_{hall.}$ is the reduction from μ_{all} to μ_{clean}. Shown only for conditions with validated flag precision.

After false-positive correction, 36.9% of Wiki's completions include fabricated laws, policies, or officials

Across prompts, Wiki's bias score tracks its hallucination rate: $r = 0.94$

Three possible, non-exclusive explanations:

1. **Overfitting with excess training steps.** 50 epochs over 10,000 articles (15,625 steps); final loss 0.032 indicates near complete memorization.
2. **Hallucination-driven score inflation.** The judge flags hallucination in 44.3% of Wiki's completions, including every Climate and Immigration completion. Correcting for the flag's false-positive rate, its mean drops from 2.153 to 1.484 (31%); the other validated conditions (GT, GT-HB) move by at most 0.077. Whether Wiki ranks above or below GT-HB after correction depends on topic weighting (pooled 1.484 vs. 1.686; equal-weighted 1.860 vs. 1.693).
3. **Amplification of pre-existing bias.** Wikipedia's own skews (domain coverage, editor demographics) plausibly shape inherited associations that fine-tuning intensifies. Most defensibly these interact: overfitting amplifies a weak immigration signal, and the base model's pro-immigration association ($d = +0.911$) supplies the fake legislative content.

6. Takeaways

- Partisan news (GT) transferred bias on specific topics; the most biased sources (GT-HB) strengthen and broaden it, surviving hallucination cleaning (1.686 > 1.439); near-neutral PS reduced bias.
- **For Llama 3.2 3B Instruct under our fine-tuning setup, corpus-level bias scores are not reliable predictors of fine-tuned model bias.** Corpora that appear neutral on the surface can still produce fine-tuned models with high bias scores when training develops into memorization.
- **In the same setup, LLM-as-Judge evaluations of bias risk conflating factual accuracy with actual bias in models that hallucinate more.**

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